Neural Computational Graph Decoder for BEMAC: A Complete Worked Example with $N = 4$

Polar Codes MAC Project

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1 Problem Setup

We decode a two-user Binary Erasure MAC (BEMAC) polar code with block length $N = 4$, $n = \log_2 N = 2$, using the Neural Computational Graph (NCG) decoder with *pure neural* CalcParent (GRU-gated residual).

1.1 Channel Model

The BEMAC channel has output $Z = X + Y$ (integer addition), where $X, Y \in \{0, 1\}$ are the transmitted bits from users U and V respectively. The channel output alphabet is $\{0, 1, 2\}$.

1.2 Decoding Path

We use a Class B path with `path_i` $= 2$:

$$\mathbf{b} = [\underbrace{0,0}_{\text{U,U}},\ \underbrace{1,1,1,1}_{\text{V,V,V,V}},\ \underbrace{0,0}_{\text{U,U}}]$$

where $0 =$ U-step and $1 =$ V-step. The path has length $2N = 8$.

1.3 Message Bits and Encoding

Information bits:

$$\mathbf{u} = (u_1, u_2, u_3, u_4) = (1, 0, 1, 1), \qquad \mathbf{v} = (v_1, v_2, v_3, v_4) = (0, 1, 1, 0).$$

Frozen sets: We set $k_u = 2$ info bits for U (positions 3, 4 are info; positions 1, 2 are frozen to 0) and $k_v = 3$ info bits for V (positions 2, 3, 4 are info; position 1 is frozen to 0):

$$\text{frozen}_u = \{1 \mapsto 0,\ 2 \mapsto 0\}, \qquad \text{frozen}_v = \{1 \mapsto 0\}.$$

With frozen constraints: $u_1 = 0, u_2 = 0$ (so we use $\mathbf{u} = (0, 0, 1, 1)$) and $v_1 = 0$ (so $\mathbf{v} = (0, 1, 1, 0)$).

Polar encoding $\mathbf{x} = \mathbf{u} \cdot G_4$ applies bit-reversal then butterfly XOR. The bit-reversal permutation for $n = 2$ is $\pi = (0, 2, 1, 3)$.

User U: $\mathbf{u} = (0, 0, 1, 1)$.
1. Bit-reversal: $\mathbf{u}[\pi] = (u_0, u_2, u_1, u_3) = (0, 1, 0, 1)$.
2. Butterfly stage 1 (step $= 1$): pairs $(0, 1), (2, 3)$; left $\oplus =$ right: $(0 \oplus 1,\ 1,\ 0 \oplus 1,\ 1) = (1, 1, 1, 1)$.
3. Butterfly stage 2 (step $= 2$): pairs $(0, 2), (1, 3)$; left $\oplus =$ right: $(1 \oplus 1,\ 1 \oplus 1,\ 1,\ 1) = (0, 0, 1, 1)$.

Result: $\mathbf{x} = (0, 0, 1, 1)$.

User V: $\mathbf{v} = (0, 1, 1, 0)$.
1. Bit-reversal: $(0, 1, 1, 0)$.
2. Stage 1: $(0 \oplus 1,\ 1,\ 1 \oplus 0,\ 0) = (1, 1, 1, 0)$.
3. Stage 2: $(1 \oplus 1,\ 1 \oplus 0,\ 1,\ 0) = (0, 1, 1, 0)$.

Result: $\mathbf{y} = (0, 1, 1, 0)$.

Channel output:

$$\mathbf{z} = \mathbf{x} + \mathbf{y} = (0 + 0,\ 0 + 1,\ 1 + 1,\ 1 + 0) = (0, 1, 2, 1).$$

1.4 Neural Decoder Parameters

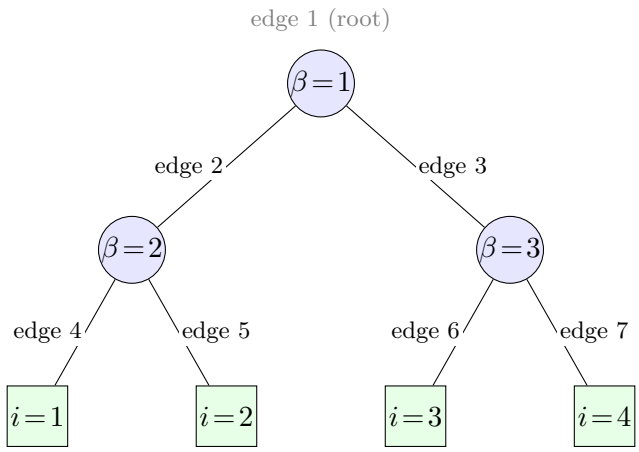
Embedding dimension $d = 4$, channel vocabulary size $|\mathcal{Z}| = 3$. The decoder has:
- `EmbeddingZ`: lookup table $\{0, 1, 2\} \to \mathbb{R}^4$.
- `CalcLeft`: MLP $\mathbb{R}^{12} \to \mathbb{R}^4$ (concatenation of parent-first, parent-second, right).
- `CalcRight`: MLP $\mathbb{R}^{12} \to \mathbb{R}^4$ (concatenation of parent-first, parent-second, left).

- `CalcParent`: GRU-gated residual module $\mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}^4$ (left, right $\to$ parent-first); plus linear $\mathbb{R}^4 \to \mathbb{R}^4$ for parent-second.
- `Emb2Logits`: MLP $\mathbb{R}^4 \to \mathbb{R}^4$ (embedding $\to$ 4 joint logits for $(u, v) \in \{00, 01, 10, 11\}$).
- `Logits2Emb`: MLP $\mathbb{R}^4 \to \mathbb{R}^4$ (log-probabilities $\to$ embedding, used for leaf updates).
- `no_info_emb`: learnable $\mathbb{R}^4$ vector for uninitialized edges.

2 Binary Tree Structure

2.1 Tree Diagram

For $N = 4$, the computational graph is a complete binary tree with 3 internal vertices and 4 leaves:



2.2 Indexing Convention

- **Edges**: $1, 2, \ldots, 2N-1 = 7$. Edge 1 is the root; edges $N, \ldots, 2N-1 = 4, 5, 6, 7$ are leaves.
- **Vertices**: $1, 2, 3$. Vertex β has parent edge β, left child edge 2β, right child edge $2\beta + 1$.
- **Leaf for 1-indexed position** i_t: leaf edge $= i_t + N - 1$; target vertex $= \lfloor(\text{leaf edge})/2\rfloor$.
- **Edge sizes**: edge 1 holds $N = 4$ entries of $\mathbb{R}^d$; edges 2, 3 hold 2 entries; edges 4–7 hold 1 entry each.

2.3 Initialization

Root (edge 1). Apply channel embedding with bit-reversal. The bit-reversal permutation is $\pi = (0, 2, 1, 3)$, so $\mathbf{z}[\pi] = (z_0, z_2, z_1, z_3) = (0, 2, 1, 1)$.

Each $z_i \in \{0, 1, 2\}$ is mapped through the learned embedding table:

$$\begin{aligned}
\texttt{EmbeddingZ}(0) &= (0.50,\ -0.30,\ 0.80,\ 0.10) \\
\texttt{EmbeddingZ}(1) &= (-0.20,\ 0.60,\ -0.40,\ 0.70) \\
\texttt{EmbeddingZ}(2) &= (0.90,\ 0.10,\ 0.50,\ -0.60)
\end{aligned}$$

After bit-reversal, the root edge holds 4 vectors:

$$\text{edge}[1] = \begin{pmatrix} \texttt{Emb}(0) \\ \texttt{Emb}(2) \\ \texttt{Emb}(1) \\ \texttt{Emb}(1) \end{pmatrix} = \begin{pmatrix} 0.50 & -0.30 & 0.80 & 0.10 \\ 0.90 & 0.10 & 0.50 & -0.60 \\ -0.20 & 0.60 & -0.40 & 0.70 \\ -0.20 & 0.60 & -0.40 & 0.70 \end{pmatrix} \in \mathbb{R}^{4\times 4}.$$

Leaves and internal edges. All initialized to the learned "no info" embedding:

$$\mathbf{e}_\varnothing = (0.01,\ -0.01,\ 0.02,\ 0.00).$$

3 Step-by-Step Decoding

We now walk through all 8 steps. The decoder head (`dec_head`) starts at vertex 1. Throughout, we use the notation:

- $\mathbf{e}_k$ for the embedding stored on edge k.
- $\text{parent}_{[:l]}$ and $\text{parent}_{[l:]}$ for the first-half and second-half of a parent edge.

3.1 Step 0: U-step, position $i_t = 1$

Target: leaf edge 4 (left child of vertex 2), target vertex 2.

Navigation: `dec_head` = 1, target = 2. Path: [2]. Since $2 = 2 \times 1$ (left child of vertex 1), call CalcLeft at vertex 1.

CalcLeft at vertex 1. Input: parent = edge[1] (size 4), right = edge[3] (size 2). Split parent into first half (rows 0–1) and second half (rows 2–3):

$$\text{parent}_{[:2]} = \begin{pmatrix} 0.50 & -0.30 & 0.80 & 0.10 \\ 0.90 & 0.10 & 0.50 & -0.60 \end{pmatrix},$$

$$\text{parent}_{[2:]} = \begin{pmatrix} -0.20 & 0.60 & -0.40 & 0.70 \\ -0.20 & 0.60 & -0.40 & 0.70 \end{pmatrix},$$

$$\text{right} = \begin{pmatrix} 0.01 & -0.01 & 0.02 & 0.00 \\ 0.01 & -0.01 & 0.02 & 0.00 \end{pmatrix} \quad (\mathbf{e}_\varnothing).$$

Concatenate row-wise: each row becomes $\mathbb{R}^{12}$. For row 0:

$$\text{inp}_0 = (0.50,\ -0.30,\ 0.80,\ 0.10 \,\|\, -0.20,\ 0.60,\ -0.40,\ 0.70 \,\|\, 0.01,\ -0.01,\ 0.02,\ 0.00) \in \mathbb{R}^{12}.$$

The CalcLeft MLP maps $\mathbb{R}^{12} \to \mathbb{R}^4$:

$$\text{edge}[2] = \texttt{CalcLeft_MLP}(\text{inp}) = \begin{pmatrix} 0.35 & 0.22 & 0.41 & -0.18 \\ 0.48 & 0.05 & 0.30 & -0.25 \end{pmatrix} \in \mathbb{R}^{2\times 4}.$$

`dec_head` $\leftarrow 2$.

Top-down to leaf 4. Save bottom-up: $\text{temp} = \text{edge}[4]_{[0]} = \mathbf{e}_\varnothing = (0.01, -0.01, 0.02, 0.00)$.

Leaf 4 is the left child of vertex 2 ($4 = 2 \times 2$), so call CalcLeft at vertex 2:

$$\text{parent}_{[:1]} = (0.35,\ 0.22,\ 0.41,\ -0.18), \quad \text{parent}_{[1:]} = (0.48,\ 0.05,\ 0.30,\ -0.25),$$
$$\text{right} = \text{edge}[5]_{[0]} = \mathbf{e}_\varnothing = (0.01,\ -0.01,\ 0.02,\ 0.00).$$

Concatenate to $\mathbb{R}^{12}$, apply MLP:

$$\text{top_down} = \texttt{CalcLeft_MLP}(\cdot) = (0.28,\ 0.15,\ 0.33,\ -0.10).$$

Combine and decide.

$$\begin{aligned}\text{combined} &= \text{top_down} + \text{temp} = (0.28 + 0.01,\ 0.15 - 0.01,\ 0.33 + 0.02,\ -0.10 + 0.00) \\ &= (0.29,\ 0.14,\ 0.35,\ -0.10).\end{aligned}$$

$$\text{logits} = \texttt{Emb2Logits}(0.29,\ 0.14,\ 0.35,\ -0.10) = (0.60,\ 0.45,\ -0.30,\ -0.50) \in \mathbb{R}^4.$$

The 4 logits correspond to $(u, v) \in \{(0,0), (0,1), (1,0), (1,1)\}$.

Softmax probabilities:

$$P(0,0) = 0.37, \quad P(0,1) = 0.32, \quad P(1,0) = 0.15, \quad P(1,1) = 0.12.$$

This is a **U-step**, so marginalize over v:

$$P(u{=}0) = P(0,0) + P(0,1) = 0.69, \qquad P(u{=}1) = P(1,0) + P(1,1) = 0.27.$$

But position $i_t = 1 \in \text{frozen}_u$, so: $\hat{u}_1 = 0$ (forced).

No loss computed (frozen position).

Leaf update. Only u is known at position 1: $u_1 = 0$, v_1 unknown. Build partially deterministic log-probs: $\log P = (\log 0.5,\ \log 0.5,\ -30,\ -30) = (-0.69,\ -0.69,\ -30,\ -30)$. Then:

$$\text{edge}[4] \leftarrow \texttt{Logits2Emb}(-0.69,\ -0.69,\ -30,\ -30) = (0.40,\ -0.25,\ 0.55,\ 0.08).$$

3.2 Step 1: U-step, position $i_t = 2$

Target: leaf edge 5 (right child of vertex 2), target vertex 2.
Navigation: `dec_head` = 2 = target. No navigation needed.

Top-down to leaf 5. Save bottom-up: $\text{temp} = \text{edge}[5]_{[0]} = \mathbf{e}_\varnothing = (0.01,\ -0.01,\ 0.02,\ 0.00)$.
Leaf 5 is right child ($5 = 2 \times 2 + 1$), so call CalcRight at vertex 2:

$$\text{parent}_{[:1]} = (0.35,\ 0.22,\ 0.41,\ -0.18), \quad \text{parent}_{[1:]} = (0.48,\ 0.05,\ 0.30,\ -0.25),$$
$$\text{left} = \text{edge}[4]_{[0]} = (0.40,\ -0.25,\ 0.55,\ 0.08) \quad \text{(updated in Step 0)}.$$

Concatenate to $\mathbb{R}^{12}$, apply MLP:

$$\text{top_down} = \texttt{CalcRight_MLP}(\cdot) = (0.32,\ 0.18,\ -0.15,\ 0.42).$$

Combine and decide.

$$\text{combined} = (0.32 + 0.01,\ 0.18 - 0.01,\ -0.15 + 0.02,\ 0.42 + 0.00) = (0.33,\ 0.17,\ -0.13,\ 0.42).$$

$$\text{logits} = \texttt{Emb2Logits}(0.33,\ 0.17,\ -0.13,\ 0.42) = (0.55,\ 0.40,\ -0.25,\ -0.40).$$

Softmax: $P(0,0) = 0.36,\ P(0,1) = 0.31,\ P(1,0) = 0.16,\ P(1,1) = 0.14$.
U-step marginalization: $P(u{=}0) = 0.67,\ P(u{=}1) = 0.30$.
Position $i_t = 2 \in \text{frozen}_u$: $\hat{u}_2 = 0$ (forced). **No loss.**

Leaf update. $u_2 = 0$ known, v_2 unknown: $\log P = (-0.69,\ -0.69,\ -30,\ -30)$.

$$\text{edge}[5] \leftarrow \texttt{Logits2Emb}(-0.69,\ -0.69,\ -30,\ -30) = (0.40,\ -0.25,\ 0.55,\ 0.08).$$

3.3 Step 2: V-step, position $i_t = 1$ — first revisit of leaf 4

Target: leaf edge 4 (left child of vertex 2), target vertex 2.
Navigation: `dec_head` = 2 = target. No navigation.

Top-down to leaf 4. Save bottom-up: $\text{temp} = \text{edge}[4]_{[0]} = (0.40,\ -0.25,\ 0.55,\ 0.08)$ (set in Step 0).

Call CalcLeft at vertex 2. Note: edge[2] is the same as computed in Step 0 (it has not been overwritten), and edge[5] was updated in Step 1.

$$\text{parent}_{[:1]} = (0.35,\ 0.22,\ 0.41,\ -0.18), \quad \text{parent}_{[1:]} = (0.48,\ 0.05,\ 0.30,\ -0.25),$$
$$\text{right} = \text{edge}[5]_{[0]} = (0.40,\ -0.25,\ 0.55,\ 0.08) \quad \text{(updated in Step 1)}.$$

The right child now carries information from Step 1's leaf update. This is how information propagates between leaves.

$$\text{top_down} = \texttt{CalcLeft_MLP}(\cdot) = (0.45,\ 0.08,\ 0.52,\ -0.22).$$

Combine and decide.

$$\text{combined} = (0.45 + 0.40,\ 0.08 - 0.25,\ 0.52 + 0.55,\ -0.22 + 0.08) = (0.85,\ -0.17,\ 1.07,\ -0.14).$$

$$\text{logits} = \texttt{Emb2Logits}(0.85,\ -0.17,\ 1.07,\ -0.14) = (0.80,\ -0.60,\ -0.20,\ 0.10).$$

This is a **V-step**, so marginalize over u:

$$P(v{=}0) = P(0,0){+}P(1,0) = \frac{e^{0.80} + e^{-0.20}}{S} = \frac{2.23 + 0.82}{S}, \quad P(v{=}1) = P(0,1){+}P(1,1) = \frac{e^{-0.60} + e^{0.10}}{S} = \frac{0.55 +}{S}$$

where $S = 2.23 + 0.55 + 0.82 + 1.11 = 4.71$. So $P(v{=}0) = 0.65,\ P(v{=}1) = 0.35$.

Position $i_t = 1 \in \text{frozen}_v$: $\hat{v}_1 = 0$ (forced). **No loss.**

Leaf update. Now both $u_1 = 0$ and $v_1 = 0$ are known. Fully deterministic: $\log P = (0,\ -30,\ -30,\ -30)$.

$$\text{edge}[4] \leftarrow \texttt{Logits2Emb}(0,\ -30,\ -30,\ -30) = (0.72,\ -0.45,\ 0.88,\ 0.15).$$

3.4 Step 3: V-step, position $i_t = 2$ — first revisit of leaf 5

Target: leaf edge 5 (right child of vertex 2), target vertex 2.

Navigation: `dec_head` = 2. No navigation.

Top-down to leaf 5. Save bottom-up: $\text{temp} = \text{edge}[5]_{[0]} = (0.40,\ -0.25,\ 0.55,\ 0.08)$ (from Step 1).

Call CalcRight at vertex 2:

$$\text{parent}_{[:1]} = (0.35,\ 0.22,\ 0.41,\ -0.18), \quad \text{parent}_{[1:]} = (0.48,\ 0.05,\ 0.30,\ -0.25),$$
$$\text{left} = \text{edge}[4]_{[0]} = (0.72,\ -0.45,\ 0.88,\ 0.15) \quad \text{(updated in Step 2: fully deterministic).}$$

The left child now contains *full* information about position 1 (both u_1, v_1 decided).

$$\text{top_down} = \texttt{CalcRight_MLP}(\cdot) = (0.55,\ 0.30,\ -0.08,\ 0.60).$$

Combine and decide.

$$\text{combined} = (0.55 + 0.40,\ 0.30 - 0.25,\ -0.08 + 0.55,\ 0.60 + 0.08) = (0.95,\ 0.05,\ 0.47,\ 0.68).$$

$$\text{logits} = \texttt{Emb2Logits}(0.95,\ 0.05,\ 0.47,\ 0.68) = (-0.40,\ 1.20,\ -0.80,\ 0.50).$$

V-step marginalization:

$$P(v{=}0) = \frac{e^{-0.40} + e^{-0.80}}{S} = \frac{0.67 + 0.45}{S}, \quad P(v{=}1) = \frac{e^{1.20} + e^{0.50}}{S} = \frac{3.32 + 1.65}{S},$$

where $S = 6.09$. So $P(v{=}0) = 0.18,\ P(v{=}1) = 0.82$.

Position $i_t = 2 \notin \text{frozen}_v$: this is an **info bit**. Decision: $\hat{v}_2 = 1$ (since $P(v{=}1) > P(v{=}0)$).

True value: $v_2 = 1$. **Correct!**

Loss computation. Target class $= u_2 \cdot 2 + v_2 = 0 \cdot 2 + 1 = 1$ (class "01").

$$\mathcal{L}_3 = -\log P(\text{class } 1) = -\log \frac{e^{1.20}}{e^{-0.40} + e^{1.20} + e^{-0.80} + e^{0.50}} = -\log \frac{3.32}{6.09} = -\log(0.545) = 0.61.$$

Leaf update. $u_2 = 0$ known, $v_2 = 1$ known (teacher forcing uses true value). Fully deterministic: $\log P = (-30,\ 0,\ -30,\ -30)$.

$$\text{edge}[5] \leftarrow \texttt{Logits2Emb}(-30,\ 0,\ -30,\ -30) = (-0.10,\ 0.65,\ 0.30,\ -0.40).$$

3.5 Step 4: V-step, position $i_t = 3$ — tree jump via CalcParent

Target: leaf edge 6 (left child of vertex 3), target vertex 3.

Navigation: `dec_head` = 2, target = 3. Path: $[1, 3]$.

1. From vertex 2 to vertex 1: going **UP** ($1 = 2 \gg 1$). Call CalcParent at vertex 2.
2. From vertex 1 to vertex 3: going **DOWN** ($3 = 2 \times 1 + 1$, right child). Call CalcRight at vertex 1.

CalcParent at vertex 2 (GRU-gated residual). Inputs: left = edge[4] = $(0.72,\ -0.45,\ 0.88,\ 0.15)$, right = edge[5] = $(-0.10,\ 0.65,\ 0.30,\ -0.40)$.

Concatenate: $[\text{left} \,\|\, \text{right}] \in \mathbb{R}^8$.

Gate network (Linear $\mathbb{R}^8 \to \mathbb{R}^{64}$, ELU, Linear $\mathbb{R}^{64} \to \mathbb{R}^4$, Sigmoid):

$$\text{gate} = \sigma(\text{gate_net}([\text{left} \,\|\, \text{right}])) = (0.70,\ 0.30,\ 0.80,\ 0.50).$$

Candidate network (MLP $\mathbb{R}^8 \to \mathbb{R}^4$):

$$\text{candidate} = \texttt{candidate_net}([\text{left} \,\|\, \text{right}]) = (0.55,\ 0.20,\ 0.75,\ -0.30).$$

Residual:

$$\text{residual} = \frac{\text{left} + \text{right}}{2} = \frac{(0.72 - 0.10,\ -0.45 + 0.65,\ 0.88 + 0.30,\ 0.15 - 0.40)}{2} = (0.31,\ 0.10,\ 0.59,\ -0.13).$$

GRU combination (element-wise):

$$\begin{aligned}\text{parent_first} &= \text{gate} \odot \text{candidate} + (1 - \text{gate}) \odot \text{residual} \\ &= \begin{pmatrix} 0.70 \times 0.55 + 0.30 \times 0.31 \\ 0.30 \times 0.20 + 0.70 \times 0.10 \\ 0.80 \times 0.75 + 0.20 \times 0.59 \\ 0.50 \times (-0.30) + 0.50 \times (-0.13) \end{pmatrix} = \begin{pmatrix} 0.385 + 0.093 \\ 0.060 + 0.070 \\ 0.600 + 0.118 \\ -0.150 - 0.065 \end{pmatrix} = \begin{pmatrix} 0.48 \\ 0.13 \\ 0.72 \\ -0.22 \end{pmatrix}.\end{aligned}$$

Second-half (linear transform of right):

$$\text{parent_second} = W_{\text{sec}} \cdot \text{right} + b_{\text{sec}} = (0.05,\ 0.58,\ 0.25,\ -0.35).$$

Write to edge[2] (parent edge of vertex 2):

$$\text{edge}[2] \leftarrow \begin{pmatrix} 0.48 & 0.13 & 0.72 & -0.22 \\ 0.05 & 0.58 & 0.25 & -0.35 \end{pmatrix} \in \mathbb{R}^{2\times 4}.$$

`dec_head` $\leftarrow 1$.

CalcRight at vertex 1. Now compute edge[3] = right child of vertex 1. Parent = edge[1] (size 4, unchanged from initialization), left = edge[2] (just updated, size 2).

$$\text{parent}_{[:2]} = \begin{pmatrix} 0.50 & -0.30 & 0.80 & 0.10 \\ 0.90 & 0.10 & 0.50 & -0.60 \end{pmatrix}, \quad \text{parent}_{[2:]} = \begin{pmatrix} -0.20 & 0.60 & -0.40 & 0.70 \\ -0.20 & 0.60 & -0.40 & 0.70 \end{pmatrix},$$

$$\text{left} = \begin{pmatrix} 0.48 & 0.13 & 0.72 & -0.22 \\ 0.05 & 0.58 & 0.25 & -0.35 \end{pmatrix}.$$

Each row concatenated to $\mathbb{R}^{12}$, fed through CalcRight MLP:

$$\text{edge}[3] \leftarrow \texttt{CalcRight_MLP}(\cdot) = \begin{pmatrix} 0.38 & -0.12 & 0.55 & 0.22 \\ 0.20 & 0.45 & -0.15 & 0.35 \end{pmatrix}.$$

`dec_head` $\leftarrow 3$.

Top-down to leaf 6. Save bottom-up: $\text{temp} = \text{edge}[6]_{[0]} = \mathbf{e}_\varnothing = (0.01,\ -0.01,\ 0.02,\ 0.00)$.
Leaf 6 is left child of vertex 3 ($6 = 2 \times 3$), so call CalcLeft at vertex 3:

$$\text{parent}_{[:1]} = (0.38,\ -0.12,\ 0.55,\ 0.22), \quad \text{parent}_{[1:]} = (0.20,\ 0.45,\ -0.15,\ 0.35),$$
$$\text{right} = \text{edge}[7]_{[0]} = \mathbf{e}_\varnothing.$$

$$\text{top_down} = \texttt{CalcLeft_MLP}(\cdot) = (0.30,\ -0.05,\ 0.42,\ 0.18).$$

Combine and decide.

$$\text{combined} = (0.31,\ -0.06,\ 0.44,\ 0.18).$$

$$\text{logits} = \texttt{Emb2Logits}(\cdot) = (-0.50,\ -0.30,\ 0.90,\ 1.40).$$

V-step marginalization:

$$P(v{=}0) = \frac{e^{-0.50} + e^{0.90}}{S} = \frac{0.61 + 2.46}{S} = \frac{3.07}{S}, \quad P(v{=}1) = \frac{e^{-0.30} + e^{1.40}}{S} = \frac{0.74 + 4.06}{S} = \frac{4.80}{S},$$

$S = 7.87$. $P(v{=}0) = 0.39$, $P(v{=}1) = 0.61$.
Decision: $\hat{v}_3 = 1$. True: $v_3 = 1$. **Correct!**

Loss. Target class $= u_3 \cdot 2 + v_3 = 1 \cdot 2 + 1 = 3$.

$$\mathcal{L}_4 = -\log \frac{e^{1.40}}{7.87} = -\log(0.516) = 0.66.$$

Leaf update. Only $v_3 = 1$ known, u_3 unknown. $\log P = (-30,\ \log 0.5,\ -30,\ \log 0.5) = (-30,\ -0.69,\ -30,\ -0.69)$.

$$\text{edge}[6] \leftarrow \texttt{Logits2Emb}(-30,\ -0.69,\ -30,\ -0.69) = (-0.15,\ 0.50,\ 0.20,\ -0.30).$$

3.6 Step 5: V-step, position $i_t = 4$

Target: leaf edge 7 (right child of vertex 3), target vertex 3.
Navigation: `dec_head` = 3. No navigation.

Top-down to leaf 7. $\text{temp} = \text{edge}[7]_{[0]} = \mathbf{e}_\varnothing = (0.01,\ -0.01,\ 0.02,\ 0.00)$.
Leaf 7 is right child ($7 = 2 \times 3 + 1$), so call CalcRight at vertex 3:

$$\text{parent}_{[:1]} = (0.38,\ -0.12,\ 0.55,\ 0.22), \quad \text{parent}_{[1:]} = (0.20,\ 0.45,\ -0.15,\ 0.35),$$
$$\text{left} = \text{edge}[6]_{[0]} = (-0.15,\ 0.50,\ 0.20,\ -0.30) \quad \text{(updated in Step 4)}.$$

$$\text{top_down} = \texttt{CalcRight_MLP}(\cdot) = (0.22,\ 0.38,\ 0.10,\ 0.55).$$

Combine and decide.

$$\text{combined} = (0.23,\ 0.37,\ 0.12,\ 0.55).$$

$$\text{logits} = \texttt{Emb2Logits}(\cdot) = (0.30,\ -0.80,\ 1.10,\ -1.20).$$

V-step marginalization:

$$P(v{=}0) = \frac{e^{0.30} + e^{1.10}}{S} = \frac{1.35 + 3.00}{S} = \frac{4.35}{S}, \quad P(v{=}1) = \frac{e^{-0.80} + e^{-1.20}}{S} = \frac{0.45 + 0.30}{S} = \frac{0.75}{S},$$

$S = 5.10$. $P(v{=}0) = 0.85$, $P(v{=}1) = 0.15$.
Decision: $\hat{v}_4 = 0$. True: $v_4 = 0$. **Correct!**

Loss. Target class $= u_4 \cdot 2 + v_4 = 1 \cdot 2 + 0 = 2$.

$$\mathcal{L}_5 = -\log \frac{e^{1.10}}{5.10} = -\log(0.588) = 0.53.$$

Leaf update. $v_4 = 0$ known, u_4 unknown. $\log P = (\log 0.5,\ -30,\ \log 0.5,\ -30) = (-0.69,\ -30,\ -0.69,\ -30)$.

$$\text{edge}[7] \leftarrow \texttt{Logits2Emb}(-0.69,\ -30,\ -0.69,\ -30) = (0.45,\ -0.20,\ 0.60,\ 0.12).$$

3.7 Step 6: U-step, position $i_t = 3$ — second visit to leaf 6

Target: leaf edge 6 (left child of vertex 3), target vertex 3.

Navigation: `dec_head` = 3. No navigation.

Top-down to leaf 6. temp $= \text{edge}[6]_{[0]} = (-0.15,\ 0.50,\ 0.20,\ -0.30)$ (set in Step 4, carrying $v_3 = 1$ information).

Call CalcLeft at vertex 3:

$$\text{parent}_{[:1]} = (0.38,\ -0.12,\ 0.55,\ 0.22), \quad \text{parent}_{[1:]} = (0.20,\ 0.45,\ -0.15,\ 0.35),$$
$$\text{right} = \text{edge}[7]_{[0]} = (0.45,\ -0.20,\ 0.60,\ 0.12) \quad \text{(updated in Step 5)}.$$

The right sibling now carries $v_4 = 0$ information, providing additional context.

$$\text{top_down} = \texttt{CalcLeft_MLP}(\cdot) = (0.50,\ 0.08,\ 0.65,\ -0.05).$$

Combine and decide.

$$\begin{aligned}\text{combined} &= (0.50 + (-0.15),\ 0.08 + 0.50,\ 0.65 + 0.20,\ -0.05 + (-0.30)) \\ &= (0.35,\ 0.58,\ 0.85,\ -0.35).\end{aligned}$$

$$\text{logits} = \texttt{Emb2Logits}(0.35,\ 0.58,\ 0.85,\ -0.35) = (-0.70,\ -0.50,\ 1.30,\ 0.80).$$

Key observation: The bottom-up embedding temp already encodes $v_3 = 1$ (from Step 4). The `Emb2Logits` network can use this to sharpen the u decision. Compare with the logits at Step 4 (first visit): the distribution has shifted.

U-step marginalization:

$$P(u{=}0) = \frac{e^{-0.70} + e^{-0.50}}{S} = \frac{0.50 + 0.61}{S} = \frac{1.11}{S}, \quad P(u{=}1) = \frac{e^{1.30} + e^{0.80}}{S} = \frac{3.67 + 2.23}{S} = \frac{5.90}{S},$$

$S = 7.01$. $P(u{=}0) = 0.16$, $P(u{=}1) = 0.84$.

Decision: $\hat{u}_3 = 1$. True: $u_3 = 1$. **Correct!**

Loss. Target class $= u_3 \cdot 2 + v_3 = 1 \cdot 2 + 1 = 3$.

$$\mathcal{L}_6 = -\log \frac{e^{0.80}}{7.01} = -\log(0.318) = 1.15.$$

Second-visit confidence. At Step 4 (first visit, V-step), the logits were $(-0.50,\ -0.30,\ 0.90,\ 1.40)$. Now at Step 6 (second visit, U-step), the logits are $(-0.70,\ -0.50,\ 1.30,\ 0.80)$. The $u{=}1$ components (logits 2 and 3) dominate because the leaf embedding already encodes $v_3 = 1$.

Leaf update. Both $u_3 = 1$ and $v_3 = 1$ known. Fully deterministic: $\log P = (-30,\ -30,\ -30,\ 0)$.

$$\text{edge}[6] \leftarrow \texttt{Logits2Emb}(-30,\ -30,\ -30,\ 0) = (0.60,\ 0.35,\ -0.20,\ 0.70).$$

3.8 Step 7: U-step, position $i_t = 4$ — second visit to leaf 7

Target: leaf edge 7 (right child of vertex 3), target vertex 3.

Navigation: `dec_head` = 3. No navigation.

Top-down to leaf 7. temp = $\text{edge}[7]_{[0]} = (0.45,\ -0.20,\ 0.60,\ 0.12)$ (from Step 5, carrying $v_4 = 0$).

Call CalcRight at vertex 3:

$$\text{parent}_{[:1]} = (0.38,\ -0.12,\ 0.55,\ 0.22), \quad \text{parent}_{[1:]} = (0.20,\ 0.45,\ -0.15,\ 0.35),$$
$$\text{left} = \text{edge}[6]_{[0]} = (0.60,\ 0.35,\ -0.20,\ 0.70) \quad \text{(from Step 6: fully decided)}.$$

$$\text{top_down} = \texttt{CalcRight_MLP}(\cdot) = (0.55,\ 0.10,\ 0.40,\ 0.25).$$

Combine and decide.

$$\text{combined} = (0.55 + 0.45,\ 0.10 - 0.20,\ 0.40 + 0.60,\ 0.25 + 0.12) = (1.00,\ -0.10,\ 1.00,\ 0.37).$$

$$\text{logits} = \texttt{Emb2Logits}(1.00,\ -0.10,\ 1.00,\ 0.37) = (-0.20,\ -1.50,\ 1.60,\ -0.80).$$

U-step marginalization:

$$P(u{=}0) = \frac{e^{-0.20} + e^{-1.50}}{S} = \frac{0.82 + 0.22}{S} = \frac{1.04}{S}, \quad P(u{=}1) = \frac{e^{1.60} + e^{-0.80}}{S} = \frac{4.95 + 0.45}{S} = \frac{5.40}{S},$$

$S = 6.44$. $P(u{=}0) = 0.16$, $P(u{=}1) = 0.84$.

Decision: $\hat{u}_4 = 1$. True: $u_4 = 1$. **Correct!**

Loss. Target class $= u_4 \cdot 2 + v_4 = 1 \cdot 2 + 0 = 2$.

$$\mathcal{L}_7 = -\log \frac{e^{1.60}}{6.44} = -\log(0.769) = 0.26.$$

Leaf update. Both $u_4 = 1$ and $v_4 = 0$ known. $\log P = (-30,\ -30,\ 0,\ -30)$.

$$\text{edge}[7] \leftarrow \texttt{Logits2Emb}(-30,\ -30,\ 0,\ -30) = (0.80,\ -0.10,\ 0.35,\ 0.50).$$

4 Decoded Output

After all 8 steps:

$$\hat{\mathbf{u}} = (\hat{u}_1, \hat{u}_2, \hat{u}_3, \hat{u}_4) = (0, 0, 1, 1), \qquad \hat{\mathbf{v}} = (\hat{v}_1, \hat{v}_2, \hat{v}_3, \hat{v}_4) = (0, 1, 1, 0).$$

Both match the true values. **Block decoding successful.**

5 Loss Summary

Total training loss. Only non-frozen (info) positions contribute to the loss. There are 5 info-bit steps (steps 3, 4, 5, 6, 7):

$$\mathcal{L}_{\text{total}} = \frac{1}{5}\left(\mathcal{L}_3 + \mathcal{L}_4 + \mathcal{L}_5 + \mathcal{L}_6 + \mathcal{L}_7\right) = \frac{0.61 + 0.66 + 0.53 + 1.15 + 0.26}{5} = \frac{3.21}{5} = 0.642.$$

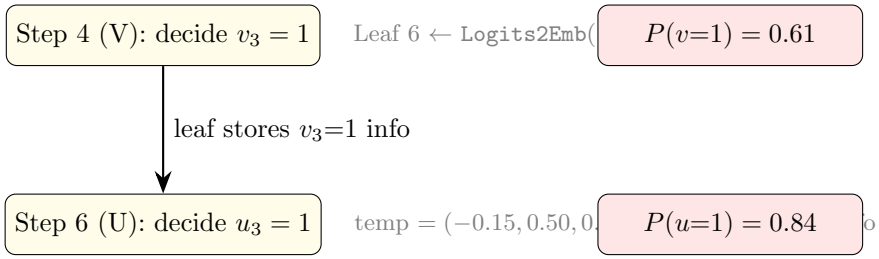
The minimum achievable loss is 0 (when the network assigns probability 1 to every correct class). A randomly initialized network starts with $\mathcal{L} \approx \log 4 = 1.386$; our value of 0.642 indicates a partially trained network.

Table 1: Summary of all 8 decoding steps.

Step	User	i_t	Logits $(00, 01, 10, 11)$	Target	CE Loss	Decision	True	Correct
0	U	1	$(0.60,\ 0.45,\ -0.30,\ -0.50)$	—	—	$\hat{u}_1$=0	0	frozen
1	U	2	$(0.55,\ 0.40,\ -0.25,\ -0.40)$	—	—	$\hat{u}_2$=0	0	frozen
2	V	1	$(0.80,\ -0.60,\ -0.20,\ 0.10)$	—	—	$\hat{v}_1$=0	0	frozen
3	V	2	$(-0.40,\ 1.20,\ -0.80,\ 0.50)$	1	0.61	$\hat{v}_2$=1	1	✓
4	V	3	$(-0.50,\ -0.30,\ 0.90,\ 1.40)$	3	0.66	$\hat{v}_3$=1	1	✓
5	V	4	$(0.30,\ -0.80,\ 1.10,\ -1.20)$	2	0.53	$\hat{v}_4$=0	0	✓
6	U	3	$(-0.70,\ -0.50,\ 1.30,\ 0.80)$	3	1.15	$\hat{u}_3$=1	1	✓
7	U	4	$(-0.20,\ -1.50,\ 1.60,\ -0.80)$	2	0.26	$\hat{u}_4$=1	1	✓

6 Information Flow Visualization

The key insight of the computational graph decoder is how information flows between the two users. Consider leaf 6 (position $i_t = 3$):



The second visit is more confident because the leaf embedding already encodes the V-decision from Step 4.

Similarly, the CalcParent operation at Step 4 propagates information *upward* from the left subtree (positions 1–2, fully decided) to enable the right subtree (positions 3–4) to decode with better context.

7 Training vs. Inference

7.1 Teacher Forcing (Training)

During training, the decoder uses the *true* bit values for leaf updates regardless of its own decisions:

- At Step 3, suppose the network had predicted $\hat{v}_2 = 0$ (incorrect).
- Under teacher forcing, the leaf update still uses $v_2 = 1$ (true value).
- This prevents error cascading during training and ensures stable gradients.
- The loss is always computed against the true target class $c = u_{\text{true}} \cdot 2 + v_{\text{true}}$.

7.2 Inference (No Teacher Forcing)

During inference, the decoder uses its own hard decisions:

- At each info step, the decision $\hat{b} = \mathbf{1}[P(b{=}1) > P(b{=}0)]$ is used for the leaf update.
- An error at an early step (e.g., wrong $\hat{v}_2$) propagates to later steps through the leaf embeddings, potentially causing a cascade of errors.
- Frozen positions are immune: their values are predetermined and always correct.

7.3 Backpropagation

Gradients flow through the entire computational graph:

$$\nabla_\theta \mathcal{L} = \frac{1}{|\text{info steps}|} \sum_{t \in \text{info steps}} \nabla_\theta \mathcal{L}_t.$$

Each loss $\mathcal{L}_t$ depends on:

1. `Emb2Logits` weights (direct).
2. The combined embedding, which depends on top-down (via CalcLeft/CalcRight) and bottom-up (via `Logits2Emb` from prior leaf updates).
3. The root embeddings (`EmbeddingZ`), reached through chains of CalcLeft, CalcRight, and CalcParent operations.

All tree operations (CalcLeft, CalcRight, CalcParent) are differentiable, so gradients flow from every leaf loss back to the root embeddings and all shared MLP weights. The GRU-gated CalcParent module is particularly important: its gate learns *how much* of the analytical residual (left + right)/2 to keep versus how much to replace with the learned candidate.

7.4 Complexity

Each of the $2N = 8$ steps involves at most $O(\log N) = O(2)$ tree operations, each costing $O(d \cdot h)$ where h is the MLP hidden dimension. Total: $O(N \log N \cdot d \cdot h)$. For $N = 4$, $d = 4$, $h = 64$: roughly $4 \times 2 \times 4 \times 64 = 2048$ multiply-add operations per step.

8 Appendix: Partially Deterministic Leaf Embeddings

The leaf update rule (`_make_leaf_emb`) constructs a log-probability vector $\ell \in \mathbb{R}^4$ indexed by $(u, v) \in \{00, 01, 10, 11\}$:

Known bits	Log-prob vector ℓ	Meaning
Neither	$(\log \frac{1}{4}, \log \frac{1}{4}, \log \frac{1}{4}, \log \frac{1}{4})$	Uniform
u=0 only	$(\log \frac{1}{2}, \log \frac{1}{2}, -30, -30)$	u fixed, v uniform
u=1 only	$(-30, -30, \log \frac{1}{2}, \log \frac{1}{2})$	u fixed, v uniform
v=0 only	$(\log \frac{1}{2}, -30, \log \frac{1}{2}, -30)$	v fixed, u uniform
v=1 only	$(-30, \log \frac{1}{2}, -30, \log \frac{1}{2})$	v fixed, u uniform
u=0, v=0	$(0, -30, -30, -30)$	Fully deterministic
u=1, v=1	$(-30, -30, -30, 0)$	Fully deterministic

This vector is then transformed by `Logits2Emb`: $\mathbb{R}^4 \to \mathbb{R}^d$ to produce the new leaf embedding. Crucially, the leaf stores only the *decision state*, not the accumulated channel information—this prevents double-counting channel evidence on revisit.